



POLITECNICO DI TORINO

Corso di Laurea in Cybersecurity

Tesi di Laurea

Iterative Approach for Network Security Automation

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Capitolo 1

Introduction

1.1 Context

Modern enterprise networks are characterized by growing complexity: hundreds of interconnected nodes, heterogeneous services, and security policies that must be consistently enforced across distributed infrastructure. Packet-filtering firewalls represent the primary mechanism for enforcing such policies, selectively allowing or blocking traffic flows based on predefined rules. However, as networks scale in size and dynamism, manually configuring firewalls becomes error-prone, time-consuming, and increasingly difficult to reason about in terms of formal correctness. An automated approach is therefore necessary — one that can guarantee both the correctness of the generated configuration and its optimality with respect to resource usage. It is in this context that VEREFOO was developed: a tool for the automated, formally correct allocation of firewalls within a network topology and the automatic generation of their filtering rules.

1.2 VEREFOO

1.2.1 Overview

VEREFOO (VERification REasoning on Firewall Optimization) is a tool designed to automate the allocation and configuration of packet-filtering firewalls in a network. It takes as input a network topology, a set of Network Security Requirements (NSRs) and a Firewall Allocation Schema (FAS), which defines the candidate positions in the topology where firewalls may be placed. From these inputs, VEREFOO produces two outputs: an updated topology indicating which nodes have been allocated as active firewalls, and a vendor-independent configuration for each firewall specifying the filtering rules to be enforced. The tool guarantees three fundamental properties: *automation*, as the entire allocation and configuration process requires no manual intervention; *formal correctness*, as the generated configuration is proven to satisfy all specified NSRs; and *optimality*, as the number of allocated firewalls and generated rules is minimised. These guarantees are achieved by encoding the problem as a MaxSMT instance and solving it with the Z3 solver. The resulting configuration can further be translated into vendor-specific formats.

1.2.2 React-VEREFOO

In operational environments, network security policies are not static: NSRs evolve over time as new services are deployed, topology changes occur, or threat models are updated. A fundamental limitation of the monolithic approach used by VEREFOO is that any change to the NSR set triggers a full recomputation of the firewall configuration from scratch. React-VEREFOO addresses this by offering a more efficient approach in the dynamic context described above, avoiding the

need to discard a valid existing configuration when only a small portion of the requirements has changed.

React-VEREFOO takes as input both the existing firewall configuration and the updated set of security requirements, and computes only the minimal changes needed to bring the network into compliance. Rather than reconsidering the entire problem, it identifies which requirements are new, which have been removed, and which remain unchanged. Only the newly added requirements drive the reconfiguration: the approach determines which parts of the network are affected and restricts the solver to those areas, leaving the rest of the configuration untouched. The result is an updated firewall allocation and rule set that satisfies all current requirements, with the same guarantees of correctness and optimality as VEREFOO.

Experimental results show that React-VEREFOO is significantly more efficient than vanilla VEREFOO when the proportion of modified NSRs is small, as the reduced problem size leads to much shorter solving times. However, when a large fraction of the requirements changes, the benefits is reduced and React-VEREFOO's performance becomes comparable to that of the monolithic approach. This observation motivates the approach investigated in this thesis: by pushing the incremental logic to its extreme - introducing only one new NSR per iteration - the proportion of modified requirements is always minimised, potentially achieving consistent speedups regardless of the total number of requirements.

Capitolo 2

Iterative React-VEREFOO

2.1 Idea of the Solution

2.2 Implementation

2.3 Example Run

Capitolo 3

Experimental Evaluation

3.1 Test Presentation

3.2 Test Configurations

3.3 Results

3.4 Comparison and Discussion

Capitolo 4

Conclusions

Capitolo 5

Executive Summary